Aryan Alpesh Bhavsar

Computer Science & Engineering **2401201001**
Indian Institute of Technology Indore Ph.D. Student

EDUCATION

Examination	University	Institute	Year	CPI/%
Doctor of Philosophy	IIT Indore	IIT Indore	2024-Present	8.88
Graduation	GTU	GIT Gandhinagar	2020-2024	8.09
Intermediate/+2	ISC	SGVP School	2018-2020	85.80
Matriculation	ICSE	SGVP School	2017-2018	80.33

RESEARCH INTEREST

I am a second year PhD student in Computer Science and Engineering at IIT Indore, advised by Dr. Sidharth Sharma (IIT Jodhpur) and Dr. Ayan Mondal (IIT Indore), working at the intersection of AI/ML systems and data center networking. My interests center on the network fabric for large-scale distributed DNN training — high-performance transport and flow control over modern Ethernet-based RDMA fabrics (RoCE, Ultra Ethernet) — where communication increasingly bottlenecks training at scale. My broader goal is to advance networks-for-AI: enabling scalable, efficient, and sustainable AI training and inference systems in large-scale data center and cloud environments.

RESEARCH EXPERIENCE & INTERNSHIPS

- **Next-Gen AI Systems & Infrastructure Research Intern** | ***Marvell Technology*** *(06/2026 - Present)*
  - Designed a GPU-cluster network fabric design tool that generates and ranks topologies (e.g., fat-tree / rail-optimized / multi-planes variants) for large-scale AI clusters up to 1M GPUs, computing switch/link counts, cost, and power for each configuration.
  - Implementing link-layer retry (LLR) and credit-based flow control (CBFC) discrete event simulation, targeting L2 flow control for lossless scale-up AI fabrics.
- **Research Scholar** | ***Indian Institute of Technology Indore*** *(09/2024 - Present)*
  - Surveyed data center congestion patterns (microbursts, incast), multicast techniques, and low-latency Ethernet transports (RoCE, Ultra Ethernet) across distributed and geo-distributed ML training.
  - Built multiple Containerlab-based example topologies and validated network configurations.
  - Implemented a symmetric RoCE-enabled IRB L3EVPN (EVPN-VXLAN) fabric in Containerlab using SR Linux switches image. Working on extending the testbed with a centralised SDN controller and live telemetry collector.
- **Software Engineer Intern** | ***Jenex Technovation Private Limited*** *(01/2024 - 06/2024)*
  - Developed 90% of key modules for a cross-platform Flutter app.
  - Implemented secure backend features in Node.js/MongoDB, token-based authentication middleware, image upload pipeline (Multer).
  - Integrated device connectivity using Android core APIs (Bluetooth) to interface with an IoT hardware component.

PUBLICATIONS

- **ScaleAcross: Designing Multi-Data-Center Infrastructure for Geo-Distributed AI Training**.
  N Inam, **AA Bhavsar**, MT Nikhil, S Sharma. *Under review, Journal of Systems Architecture*, 2026. Preprint: arXiv:2606.12963.
  - Designed and implemented a source-port allocation scheme in the Soft-RoCE (RXE) kernel module that assigns each Queue Pair to a disjoint partition of the UDP port range (instead of hashing over the whole pool), reducing ECMP path collisions among per-QP collective subflows to improve traffic distribution under Ring-AllReduce by up to 12.5% network-wide and 13.7% at the leaf layer, while preserving standard ECMP hashing and commodity switch compatibility.

SELECTED PROJECTS

- **RoCE-enabled L3EVPN over EVPN-VXLAN in a Spine–Leaf Fabric (Containerlab + Nokia SR Linux)**
  - Implemented EVPN-VXLAN L3 connectivity with multiple VNIs- validated host reachability across leafs and segmentation across VNIs.
  - Configured IRB (Integrated Routing & Bridging) to enable L2/L3 gateway functionality and validated inter-subnet reachability.
  - Implemented and validated soft-RoCE (RDMA over Converged Ethernet) within virtual machines hosted inside containerized environments (Containerlab).
  - Designed and implemented a host-driven load balancing mechanism by modifying the RXE (Soft-RNIC) source port allocation logic, replacing QP-hashing with custom port allocation and validating improved flow distribution.

TECHNICAL SKILLS

- **Tools**: htsim, ASTRA-sim, ContainerLab, Docker, make, Nokia SR Linux (Network Configuration)
- **Programming & Scripting Languages**: C++, Shell Scripting, Python; Familiar with Java, Dart, Typescript
- **Development Technology**: MERN, Flutter
- **Language:** English, French & Hindi

POR & ACHIEVEMENTS

- **Teaching Assistant** for the course, **CS334/434/634: Wireless Networks and Applications**, conducting tutorials for the course. *(01/2026 - 05/2026)*
- **Teaching Assistant** for the course, **CS653: Programming Lab**, supervised the labs for the course. *(06/2025 - 11/2025)*
- **Teaching Assistant** for the course, **CS306: Computer Network**, supervised the labs for the course. *(01/2025 - 05/2025)*
- **AIR 2308** in **GATE CSE 2024** exam- *Marks: 51.09, Score: 606* *(02/2024)*

CONTACT INFORMATION

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